

Freight Rate Prediction: Approach and Validation

1. Split and validation approach

- validation.csv is a later period (Nov-Dec 2025) than train-test.csv (Jan-Oct 2025), so random K-fold would leak time. I used an expanding-window split: for each test month M in Jun-Oct, train on every month before M and score on M. This mimics the real task of predicting the next period from the past.
- Metrics: MAE, RMSE, MAPE and median APE. Spotter's metric is unknown, so I report several and picked the model that wins on all of them. Scores are reported on clean labels and on raw labels.

2. Data-quality issues and fixes

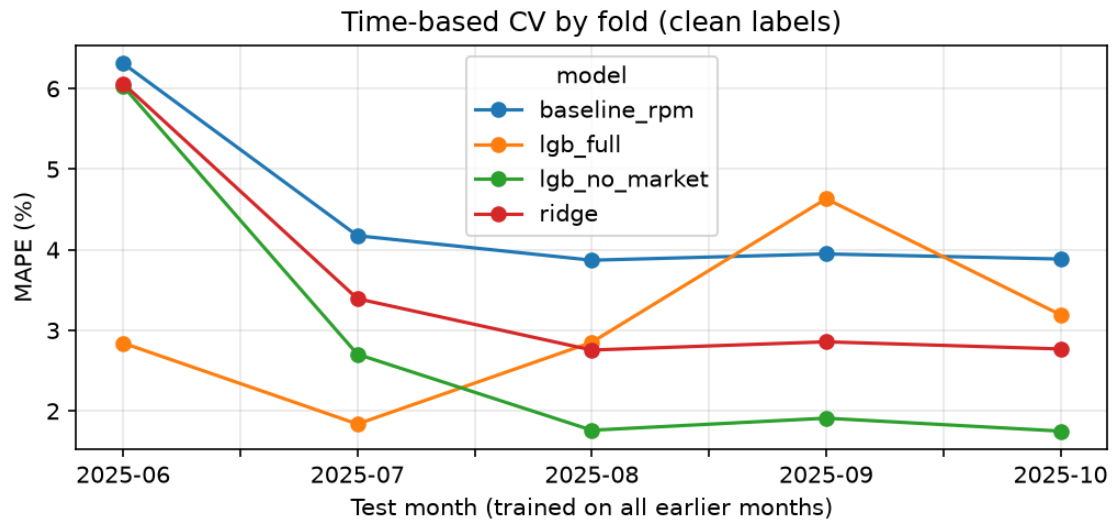
- About 1.4% of posted_rate values (677 rows) are corrupted. The corruption is near-symmetric: 340 rows sit about 3.5x above the normal rate per mile for their equipment and 337 sit about 3.6x below it, spread evenly across months and equipment types. Distance stays consistent with the coordinates on these rows, so the rate itself is wrong rather than the lane. They are detected with a deliberately low-capacity model (distance + equipment) whose log-residual exceeds 0.5, and removed from training only - never from the validation predictions, which must cover all 12,000 loads.
- Negative weights (292 train, 145 validation) are sign errors; their magnitudes match the positive distribution, so I take abs(). Missing weight (300 / 165) uses the equipment median plus an indicator.
- Missing market_index (374 / 249) is left as NaN with an indicator.

3. Model and results

Rate is driven by distance (corr 0.91) with a decreasing rate per mile, so I model $\log(\text{posted_rate})$ with LightGBM using distance, equipment, weight, origin, destination, coordinates, circuitry and day of week. Baselines: median rate per mile by equipment and distance bucket, and a Ridge model.

Scope	Model	MAE	RMSE	MAPE %	MedAPE %
clean	baseline_rpm	99.69	134.55	4.44	3.90
clean	lgb_full	71.36	96.99	3.07	2.83
clean	lgb_no_market	68.49	93.05	2.83	2.58
clean	ridge	82.47	110.34	3.57	3.22
raw	baseline_rpm	154.84	654.89	6.71	3.97
raw	lgb_full	126.70	648.43	5.31	2.87

raw	lgb_no_market	123.94	647.65	5.11	2.60
raw	ridge	137.81	651.21	5.84	3.27



Market features (market_index, quote_signal) did not improve time-based CV, so the final model excludes them. I tested them raw (3.06% MAPE) and as four stationary transforms - deviation from the daily mean, level relative to a trailing 28-day mean, and 7d/28d momentum - and none beat the 2.83% of the model without them; the best variant reached 2.95%.

The per-fold pattern explains why. Raw market features win the early folds (2.84% vs 6.03% on June) and lose the late ones (4.62% vs 1.91% on September). That tracks the market regime: market_index climbs from 0.91 in January to 1.30 in May, then reverses and falls to 0.87 by September. While the trend continues the feature helps, but gradient-boosted trees cannot extrapolate beyond their trained range, so when the regime reverses the model is confidently wrong. The signal is real but unstable, and the instability costs more than the signal earns.

The decisive folds are September and October, the closest analogues to the Nov-Dec prediction window, where excluding market features wins clearly (1.91% and 1.75% against 4.62% and 3.18%). market_index is also flat at roughly 0.92-0.95 across November and December, essentially unchanged from October, so there is no regime shift for a market feature to capture. The known failure mode is a holdout that spans a regime break, as June does.

Excluding these features also lets one model serve both the 12,000 validation loads and the December chart, where they do not exist. The raw-label RMSE is dominated by the corrupted labels, which no model can predict.

4. December prediction

The fixed lane (Lexington to Fort Wayne, 360 mi, Dry Van, 32,000 lb) is scored with the same model, so only the date changes. The curve is a weekly cycle (about \$814 to \$832, mean \$826, about \$2.29/mi), consistent with the lane's own Jan-Oct history (mean \$2.26/mi). There is no trend because the training data shows no reliable month-level effect once distance and equipment are accounted for.

